SIGCOMM 2026
DENVER

AIDA: Accelerating Root Cause Analysis for Multi-Vendor Device Failures with LLM-Powered Reasoning

<u>Haoran Xu</u>, Xuan Zeng, Xumiao Zhang, Xiaoxi Zhang,

Yang Lv, Peng Zhang, Deke Guo, Ennan Zhai

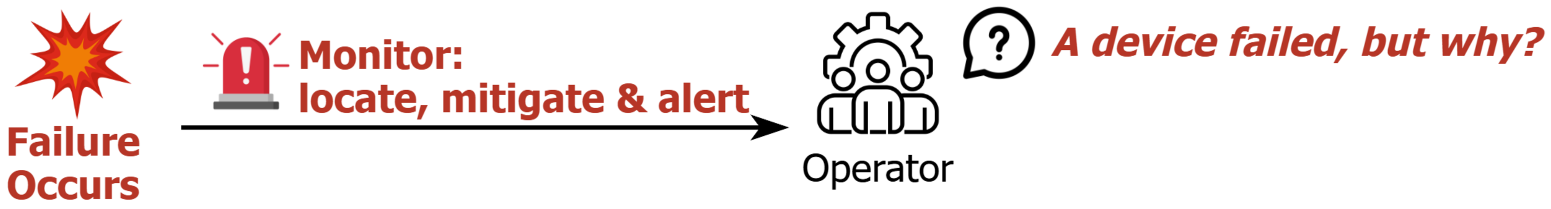
中山大學
SUN YAT-SEN UNIVERSITY

Alibaba Cloud

Network device failures: easy to detect, hard to diagnose

- Monitoring systems **quickly detect** failed devices
- But **why** did it fail?

  ⇒ critical for long-term system stability



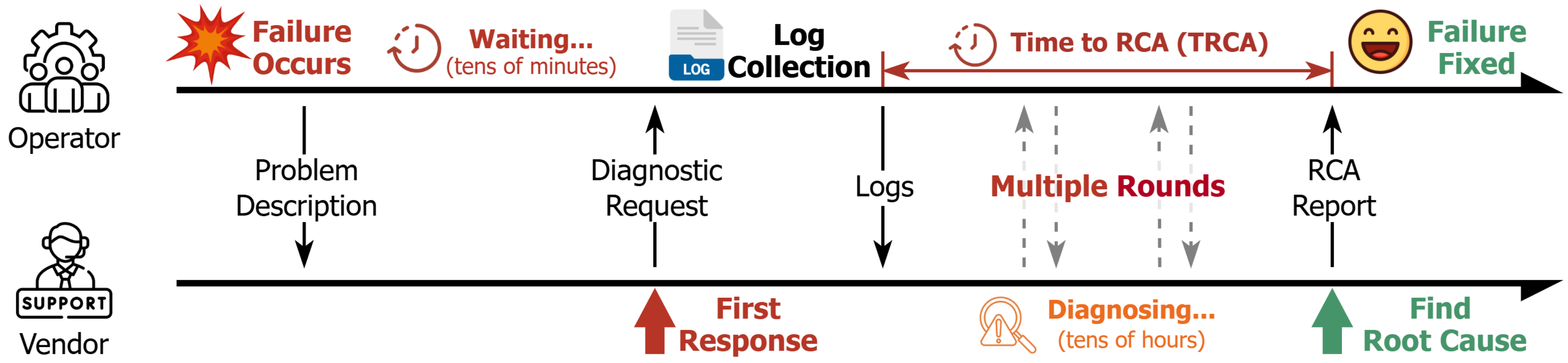
Network device failures: easy to detect, hard to diagnose

- Monitoring systems **quickly detect** failed devices
- But **why** did it fail?

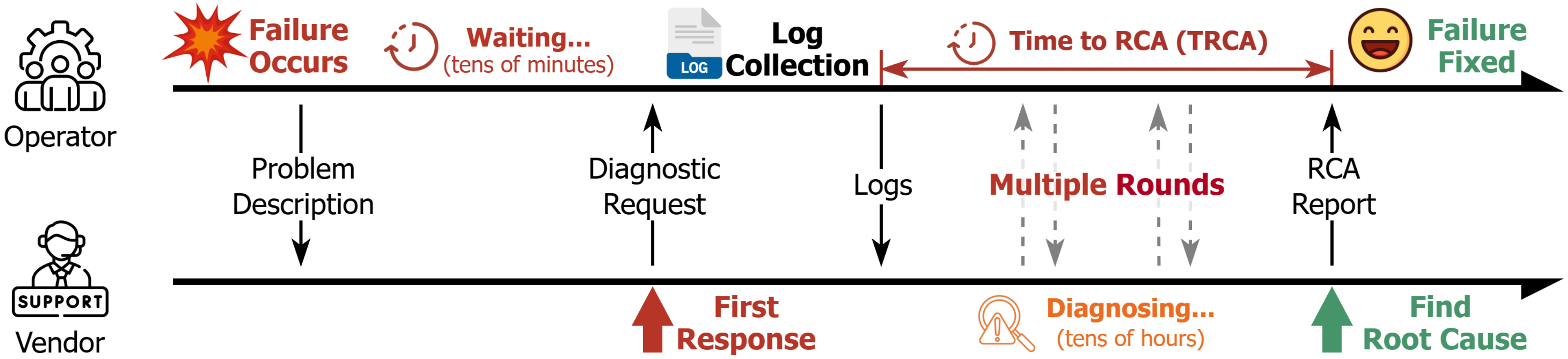
  ⇒ critical for long-term system stability

- **Multi-vendor devices,** operators lack vendor-proprietary expertise

  ⇒ must **rely on vendor experts** for **Root Cause Analysis (RCA)**

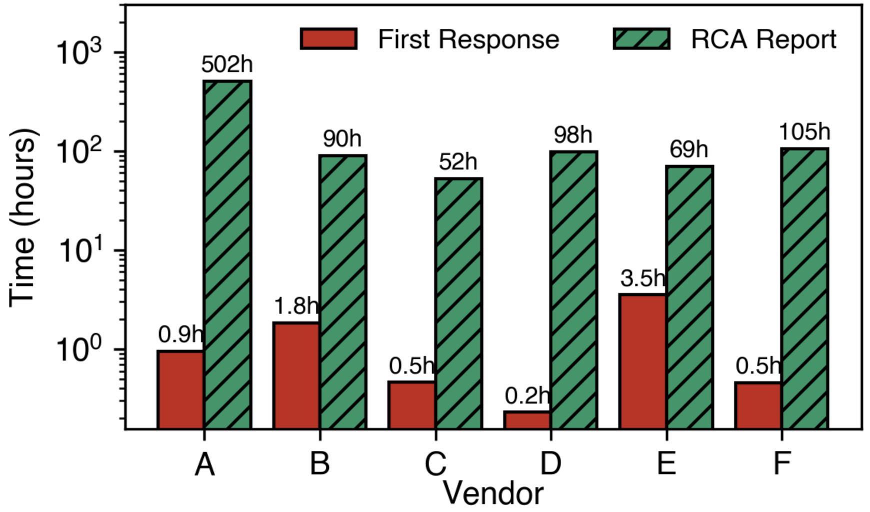


The cost: days to weeks for a single diagnosis



Before root cause is identified:

- Cannot apply appropriate fix
- Network exposed to unknown risk
- Recurring failures ⇒ degraded service



Vendor expert emails: valuable knowledge for RCA

RCA Emails

- Years of historical RCA emails encode vendor-proprietary expertise
- RCA report emails encode expert diagnostic reasoning

A sample vendor expert RCA report email

From: vendor-support@<vendor>.com • Subject: Re: Device crash on SW-DEVICE-A1

```
Hi Team,
Error log:
 Jun 7 12:28:53 ... %KERN-1-SYSTEM_MSG: RIP[]gen_pool_free+0x7c/0xb0 - kernel
Analysis:
The error was at Jun 7 12:28:53.
Logging and version output confirm the device auto-rebooted after:
 SW-DEVICE-A1# show logging log | no
 Jun 7 12:32:21 ... syslogd: restart (remote reception)
 Jun 7 12:32:22 ... %MMODE-2-MODE_CHANGED: unplanned maintenance
 Jun 7 12:32:24 ... %PLATFORM-5-PS_FOUND: Power supply 1
 Jun 7 12:32:24 ... vdc 1 state changed to create pending
 ... (boot-up messages, fan/PS status x8)
 SW-DEVICE-A1# show version
 Nexus9000, Intel(R) Xeon(R) CPU, Version: 9.2
 Last reset: Kernel Panic, Kernel uptime 0d 3h24m
The device reset reason is "Kernel Panic":
 SW-DEVICE-A1# show system reset-reason
 ------ reset reason for module 1 ------
 Reason: Kernel Panic Version: 9.2(3y)
Prior to the Kernel Panic, memory errors were reported:
 <7> EDAC sbridge MC0: HANDLING MCE MEMORY ERROR
 <7> EDAC sbridge MC0: CPU 0 Bank 10: 8c000051000
 ... (EDAC MCE MEMORY ERROR messages x5)
 <6> CCTRL PANIC DUMP
 <6> WDT last punched at 17239888
 <6> REG(0x300) = baadbeef
 <6> obfl_set_mmc_rr initialized on mmcblk0p2 blksize=512
 ... (obfl_set_mmc_rr messages x3)
 <0> nxos_panic: Kernel panic - not syncing: Fatal exception
Conclusion: Memory hardware fault. Recommend RMA replacement.
...(RMA shipping details omitted)
```

Vendor expert emails: valuable knowledge for RCA

RCA Emails

- Years of historical RCA emails encode vendor-proprietary expertise
- RCA report emails encode expert diagnostic reasoning

Characteristics

- **Implicit reasoning chain** across multiple evidence steps
- Key evidence is interleaved with noisy logs
- No explicit reasoning structure

Can we directly leverage these emails?

A sample vendor expert RCA report email

From: vendor-support@<vendor>.com • Subject: Re: Device crash on SW-DEVICE-A1

Problem Description

```
Hi Team,
Error log:
Jun 7 12:28:53 ... %KERN-1-SYSTEM_MSG: RIP[]gen_pool_free+0x7c/0xb0 - kernel
Analysis:
```

Evidence ① Confirm reboot

```
The error was at Jun 7 12:28:53.
Logging and version output confirm the device auto-rebooted after:
SW-DEVICE-A1# show logging log | no
Jun 7 12:32:21 ... syslogd: restart (remote reception)
Jun 7 12:32:22 ... %MMODE-2-MODE_CHANGED: unplanned maintenance
Jun 7 12:32:24 ... %PLATFORM-5-PS_FOUND: Power supply 1
Jun 7 12:32:24 ... vdc 1 state changed to create pending
... (boot-up messages, fan/PS status x8)
SW-DEVICE-A1# show version
Nexus9000, Intel(R) Xeon(R) CPU, Version: 9.2
Last reset: Kernel Panic, Kernel uptime 0d 3h24m
```

Evidence ② Confirm kernel panic

```
The device reset reason is "Kernel Panic":
SW-DEVICE-A1# show system reset-reason
------ reset reason for module 1 ------
Reason: Kernel Panic Version: 9.2(3y)
```

Evidence ③ Trace memory error

```
Prior to the Kernel Panic, memory errors were reported:
<7> EDAC sbridge MC0: HANDLING MCE MEMORY ERROR
<7> EDAC sbridge MC0: CPU 0 Bank 10: 8c000051000
... (EDAC MCE MEMORY ERROR messages x5)
<6> CCTRL PANIC DUMP
<6> WDT last punched at 17239888
<6> REG(0x300) = baadbeef
<6> obfl_set_mmc_rr initialized on mmcblk0p2 blksize=512
... (obfl_set_mmc_rr messages x3)
<0> nxos_panic: Kernel panic - not syncing: Fatal exception
```

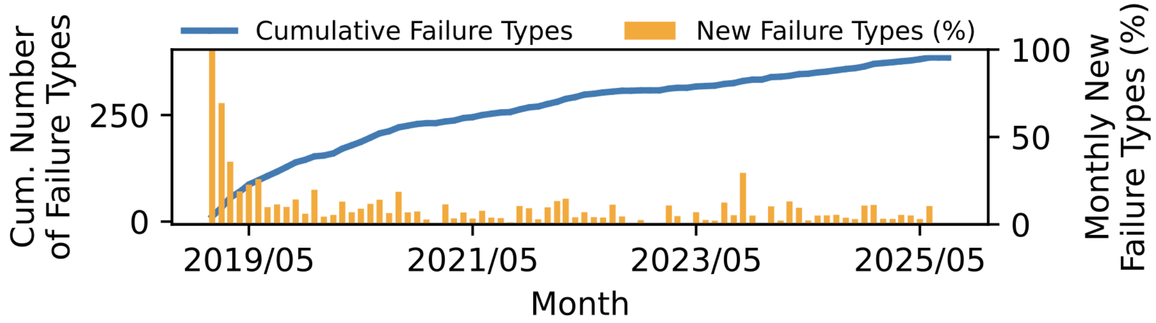
Conclusion

```
Conclusion: Memory hardware fault. Recommend RMA replacement.
...(RMA shipping details omitted)
```

Observation 1: naive RAG retrieves redundant knowledge

Naive RAG

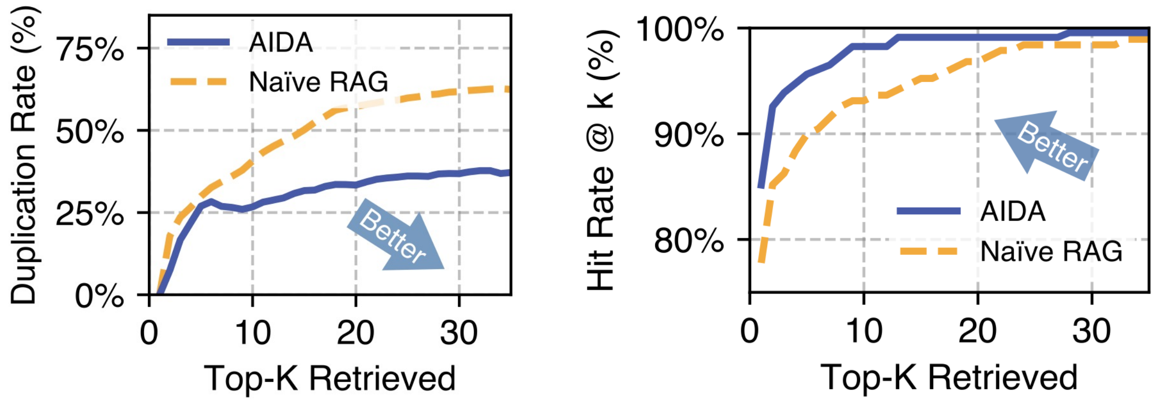
New Failure → **Retrieve Top- K similar RCA Emails** → **LLM Diagnosis**



27% of failure types occur fewer than 10 times/year

Failure knowledge is long-tailed

- A few common root causes recur frequently
- New failure types continue to emerge



More retrieval ≠ More diagnostic knowledge

Naive retrieval wastes context

- Duplicate evidence for common root causes
- Poor coverage of rare failures

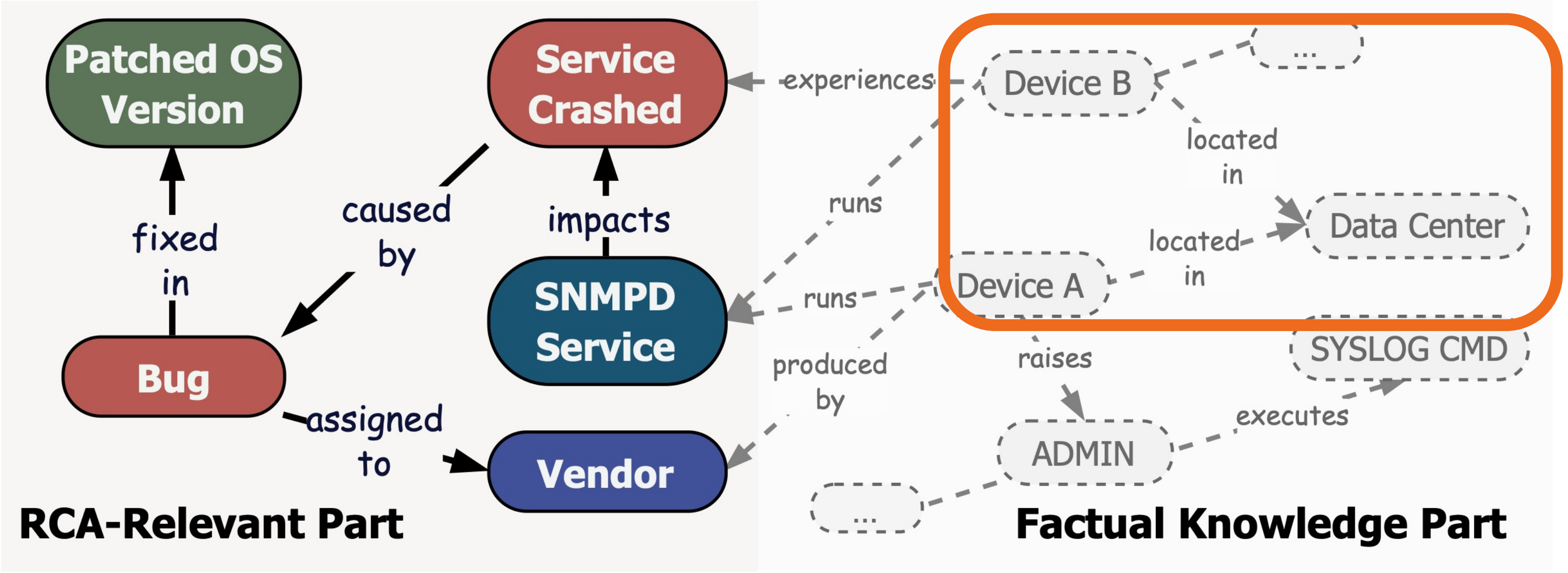
Observation 2: generic KGs miss diagnostic causality

Generic Knowledge Graph

✓ Reduce redundancy

Limitation

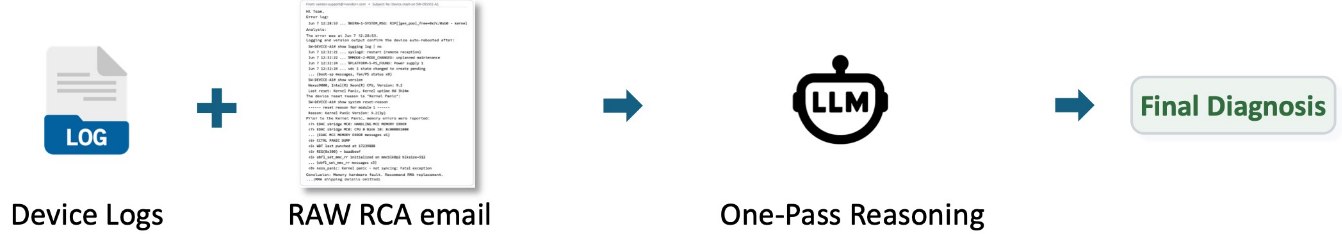
- Capture factual associations
- Miss diagnostic causality



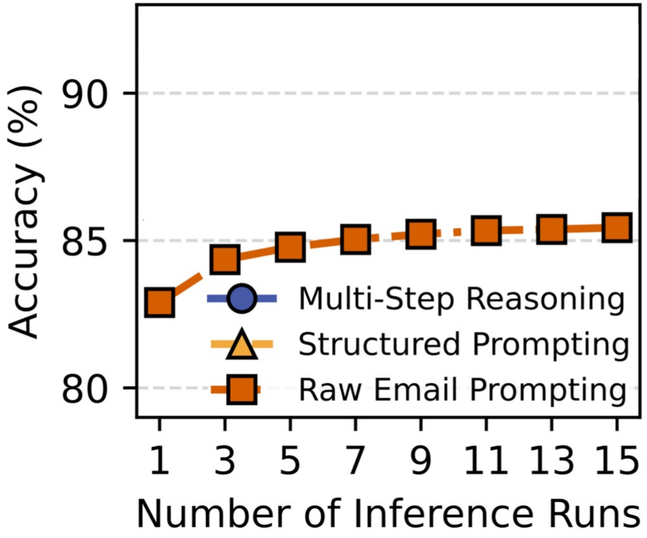
Generic KGs mix diagnostic causality with noisy factual correlations

Observation 3: better context leads to better reasoning

Raw Email Prompting *– Reason directly from* ***raw RCA emails***

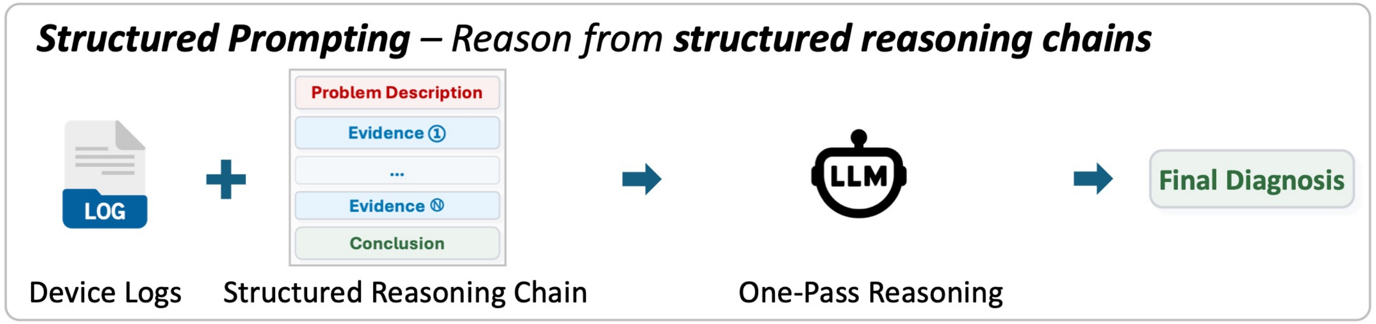


LLMs reconstruct reasoning from raw RCA emails → Low accuracy

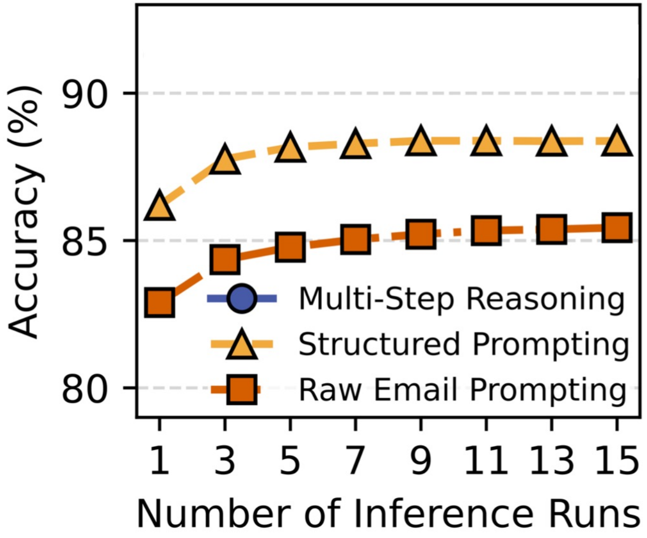


Observation 3: better context leads to better reasoning

LLMs reconstruct reasoning from raw RCA emails → Low accuracy

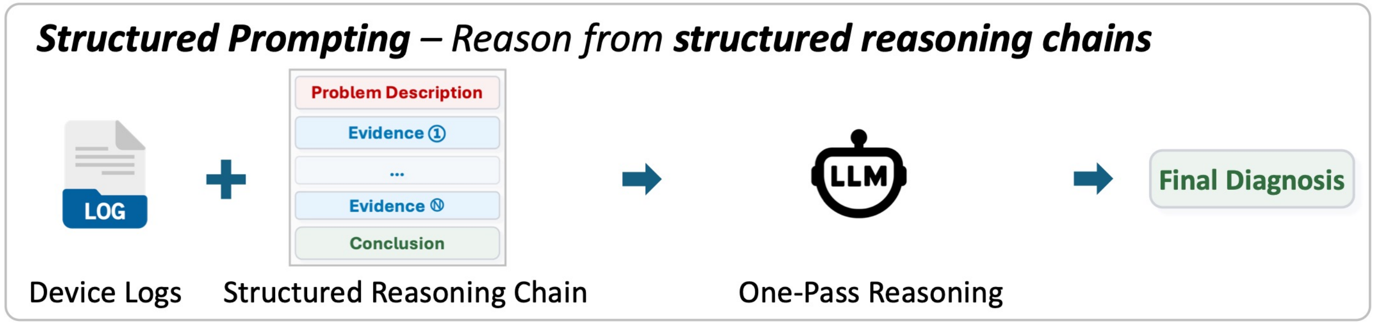


Reasoning chains structure diagnostic information → Higher accuracy

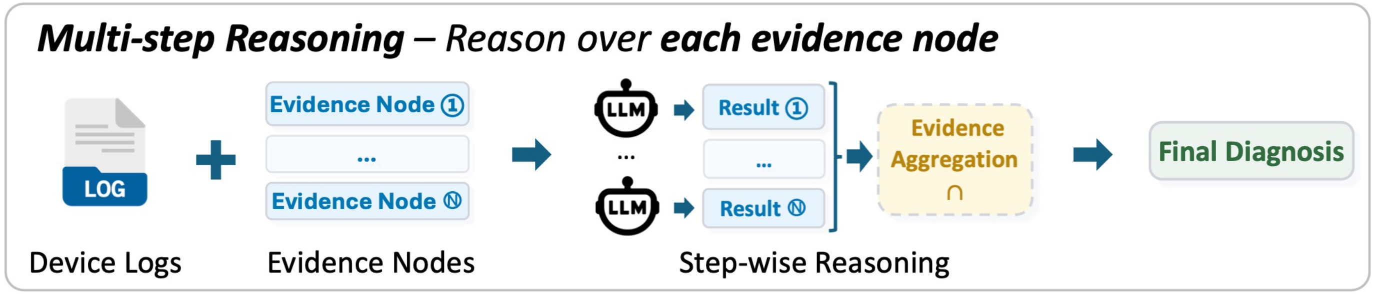


Observation 3: better context leads to better reasoning

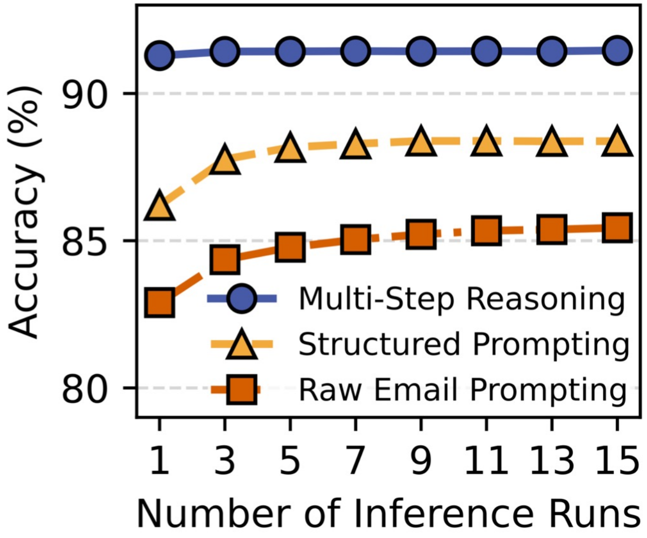
LLMs reconstruct reasoning from raw RCA emails → Low accuracy



Reasoning chains structure diagnostic information → Higher accuracy



Stepwise evidence verification → Higher, more stable accuracy



AIDA: distilling expert diagnosis into reusable reasoning

- Extract expert reasoning from historical RCA emails
- Generalize diagnostic knowledge through a causal knowledge graph
- Guide LLM reasoning with the causal KG for new failures

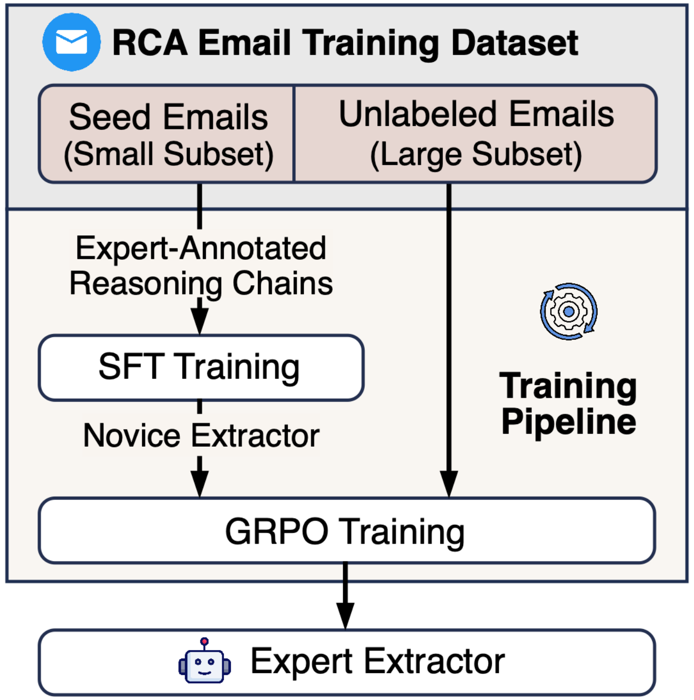
Training an expert reasoning-chain extractor

Problem: No reasoning-chain labels in historical RCA emails

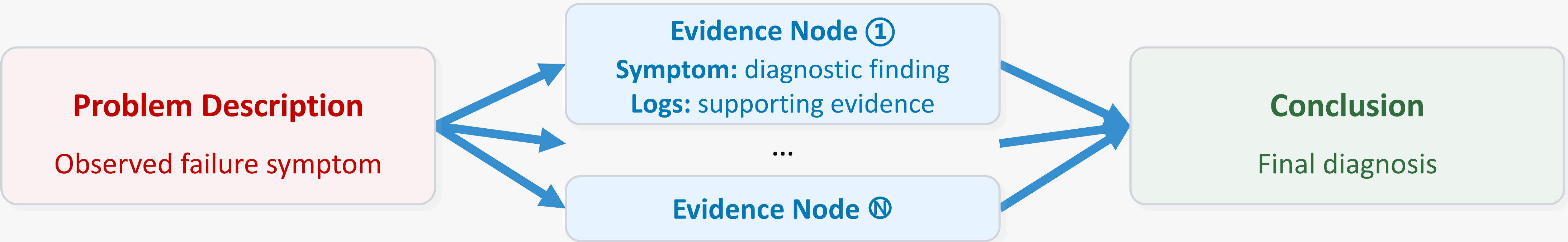
Goal: learn an extractor without large-scale annotations

RL-based LLM extractor training

- Experts annotate **~50 RCA report emails** → seed reasoning chains
- **Supervised Fine-Tuning (SFT)** bootstraps a novice extractor
- **Reinforcement learning (GRPO)** learns from the large unlabeled RCA report emails → expert extractor



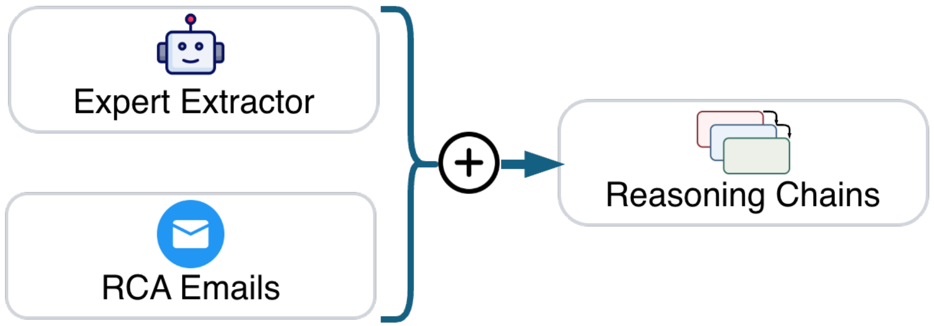
Structure of a Reasoning Chain

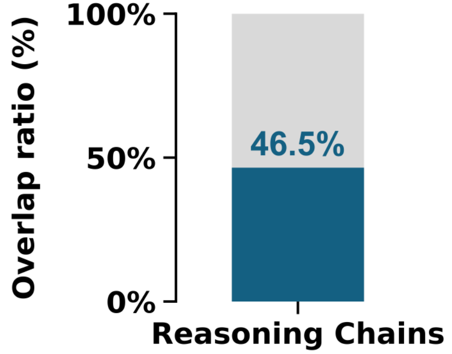


From reasoning chains to a causal RCA knowledge graph

Historical reasoning chains

- Case-specific and redundant
- 46.5% shared reasoning steps
- Continuously evolving knowledge

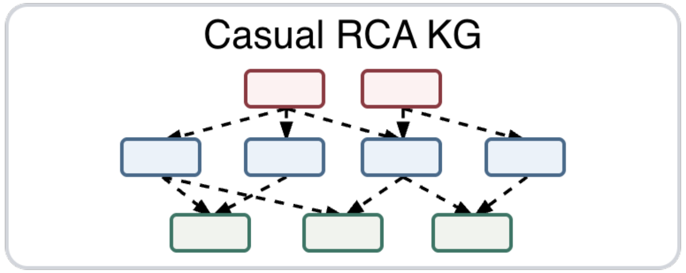




46.5% share reasoning steps

Causal RCA knowledge graph

- Consolidate shared reasoning across cases
- Generalize case-specific diagnostic knowledge
- Support efficient retrieval and updates



Building the RCA knowledge graph

Reasoning Chain C_1

Problem Description: $p^{(1)}$
`%MCE_24HR_FAIL: Correctable error threshold exceeded`

Symptom $n_1^{(1)}$
MCE correctable error threshold exceeded.
Logs
`MCE_FAIL: Module has exceeded MCE correctable threshold`

Symptom $n_2^{(1)}$
MCE diagnostic check shows **failure**.
Logs
```
Error code -> DIAG TEST SUCCESS
Consecutive failure count -> 0
Last failure reason -> Failure
```

Root Cause: Memory **hardware failure** on the device; RMA is recommended. $rc^{(1)}$

Reasoning Chain C_2

Problem Description: $p^{(2)}$
`%SYSTEM_MSG: Corrected memory error threshold exceeded`

Symptom $n_1^{(2)}$
MCE correctable error threshold exceeded.
Logs
`MCE_FAIL: Module has exceeded MCE correctable threshold`

Symptom $n_2^{(2)}$
MCE check shows **no failures**.
Logs
```
Error code -> DIAG TEST SUCCESS
Consecutive failure count -> 0
Last failure reason->No failures
```

Symptom $n_3^{(2)}$
System resource usage is normal.
Logs
```
Load avg: 0.60/0.46/0.43
CPU: 4.8% busy, 95.2% idle
Mem: 5.9G used, 16.3G total
```

Root Cause: mce_test **software bug** triggers excessive syslogs. $rc^{(2)}$

Symptom $n_2^{(1)}$
MCE diagnostic check shows **failure**.
Logs
```
Error code -> DIAG TEST SUCCESS
Consecutive failure count -> 0
Last failure reason -> Failure
```

High embedding similarity (cos ≈ 0.9)

Symptom $n_2^{(2)}$
MCE check shows **no failures**.
Logs
```
Error code -> DIAG TEST SUCCESS
Consecutive failure count -> 0
Last failure reason->No failures
```

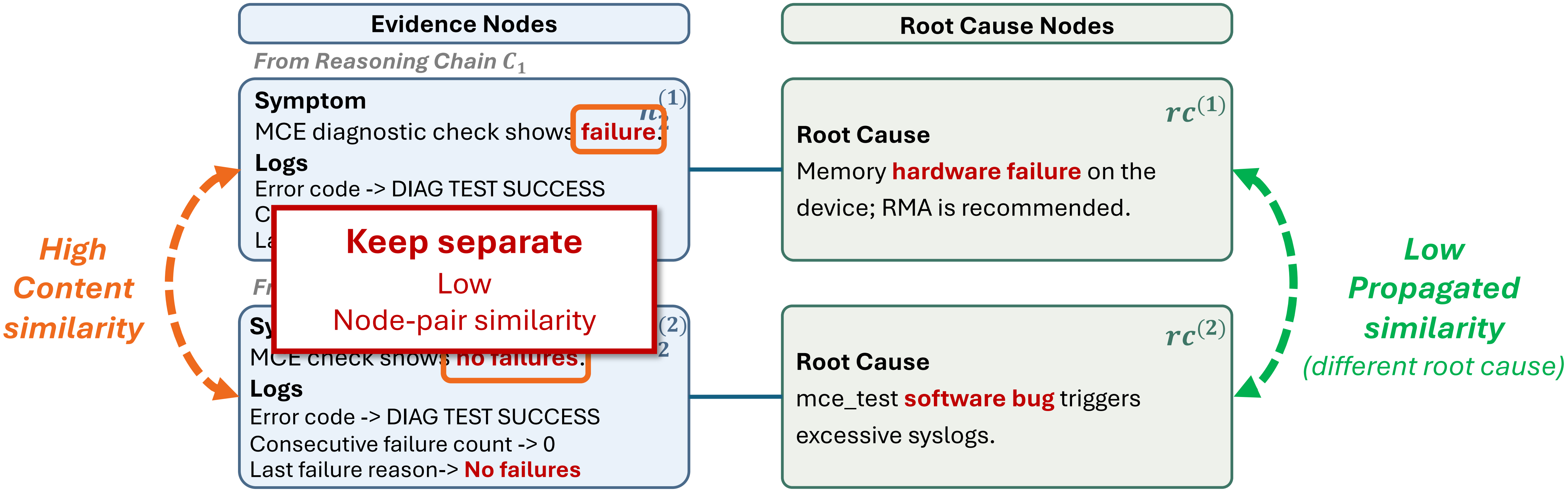
Nearly identical text, but opposite diagnostic meanings

- *Only one word differs: **"failure"** vs. **"no failure"***

✗ Incorrect merge collapses distinct diagnostic paths

Structure-aware RCA knowledge graph construction

Insight: Causal neighbors disambiguate textually similar nodes



Node-pair similarity = Content similarity + Propagated similarity

Content similarity

- Captures semantic similarity

Propagated similarity

- Captures diagnostic context from neighboring nodes

Structured RCA reasoning

New Failure → Retrieve reasoning chains → Extract evidence logs → LLM verifies evidence → Validate traces & rank root causes → Generate RCA report

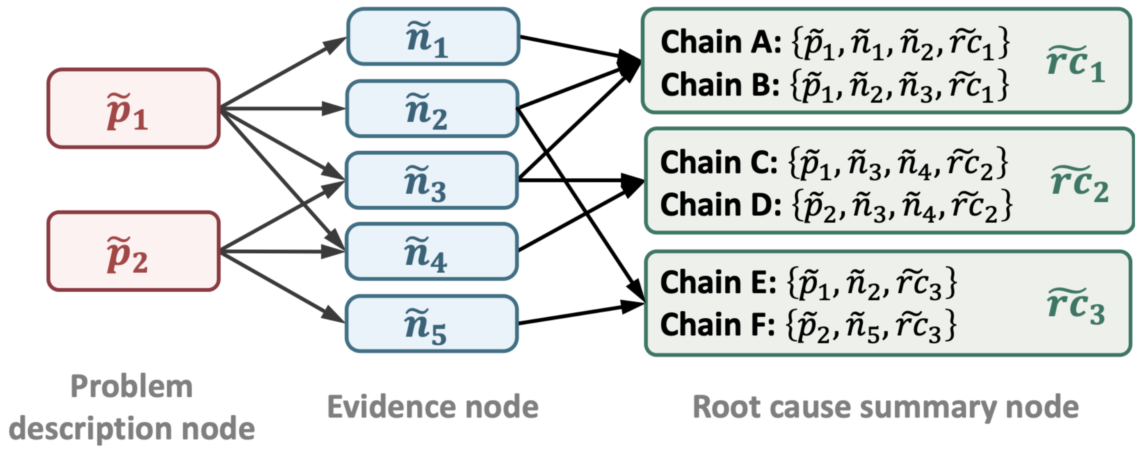
New incident

Problem Description
Physical interface: et-0/0/0:1, Enabled, Physical link is Down

Match the incident descriptions in RCA KG

Retrieved reasoning chains

Candidate diagnostic hypotheses from the RCA KG

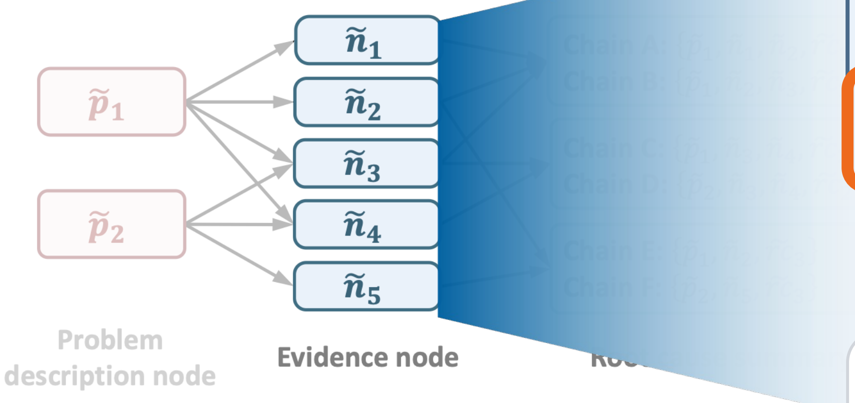


Structured RCA reasoning

New Failure → Retrieve reasoning chains → **Extract evidence logs** → LLM verifies evidence → Validate traces & rank root causes → Generate RCA report

Retrieved reasoning chains

Candidate diagnostic hypotheses from the RCA KG



LOG

Runtime logs (Raw)

MB- to GB-scale device logs

+

Symptom $\tilde{n}_1$

BFD status returned to UP after port isolation rollback

Log Templates

discriminator<*>new state<*>up

Symptom $\tilde{n}_2$

CBR port remains up during this period

Log Templates

Link-level<*>LAN-PHY<*>BPDU

Filter logs for each evidence node

Extracted logs A

22:53:22 BGP peer 100.0.0.1 ... state from OpenConfirm to Established ...

22:53:22 local discriminator: 45, new state: up, interface: et- 0/0/35.0, peer addr: 100.0.0.2 ...

22:53:22 Login failed for user 'xnetadmin' from host '100.0.0.3' ...

Extracted logs B

Physical interface et-0/0/0:1: enabled, link down...

Link-level: Ethernet, LAN-PHY, 10Gbps, no BPDU/other errors...

Physical interface et-0/0/0:2: enabled, link down...

Link-level: Ethernet, LAN-PHY, 10Gbps, no BPDU/other errors...

Physical interface et-0/0/0:3: enabled, link down...

Reasoning chain + Current incident logs → Grounded reasoning trace

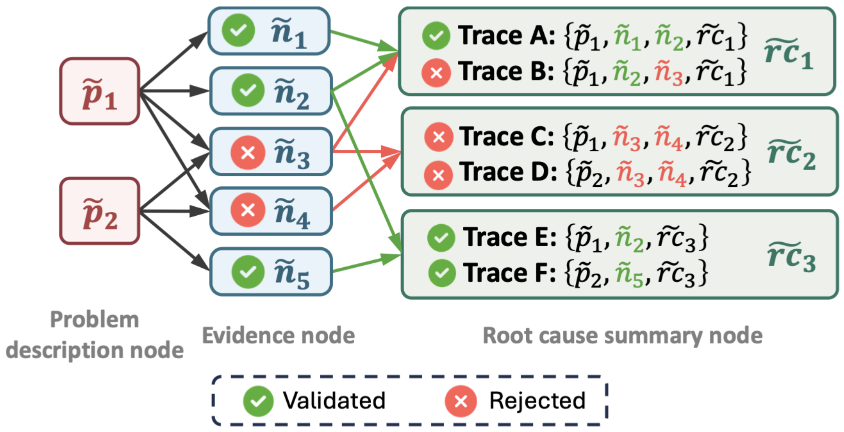
** Template-generation details are provided in our paper.*

Structured RCA reasoning

New Failure → Retrieve reasoning chains → Extract evidence logs → **LLM verifies evidence** → Validate traces & rank root causes → Generate RCA report

Verify each evidence node against filtered log snippets

Candidate reasoning traces



✓ All evidence nodes validated ⇒ Trace survives

Symptom
BFD status returned to UP after port isolation rollback

Extracted logs A
22:53:22 BGP peer 100.0.0.1 ... state from OpenConfirm to Established ...
22:53:22 local discriminator: 45, new state: up, interface: et- 0/0/35.0, peer addr: 100.0.0.2 ...
22:53:22 Login failed for user 'xnetadmin' from host '100.0.0.3' ...

Symptom
CBR port remains up during this period

Extracted logs B
Physical interface et-0/0/0:1: enabled, link down...
Link-level: Ethernet, LAN-PHY, 10Gbps, no BPDU/other errors...
Physical interface et-0/0/0:2: enabled, link down...
Link-level: Ethernet, LAN-PHY, 10Gbps, no BPDU/other errors...
Physical interface et-0/0/0:3: enabled, link down...

LLM evidence reasoning

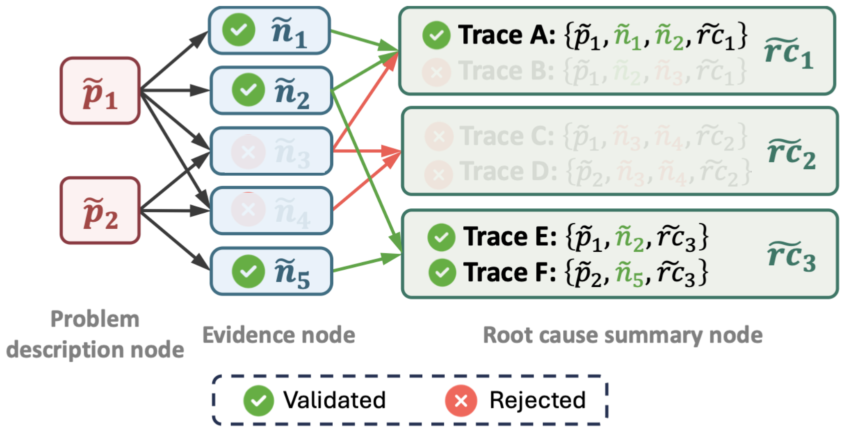
Analysis A — Validated
*... BGP peer state changed to Established ... **BFD new state: up** ... interface et-0/0/35.0 ... BFD status returned to UP ...*

Analysis B — Rejected
*... link down ... **no evidence of operational link up** ... cannot support the symptom that the CBR port remained up...*

Structured RCA reasoning

New Failure → Retrieve reasoning chains → Extract evidence logs → LLM verifies evidence → **Validate traces & rank root causes** → **Generate RCA report**

End-to-end trace validation



Trace validation

- ✓ All evidence nodes validated
- ✓ Same device/module & timeline

→

Surviving traces

- Trace A
- Trace E
- Trace F

Confidence-based root-cause ranking

Trace confidence:

- Reasoning chain reliability
- Current-log support across all evidence steps

  Weak support at any step lowers the trace confidence

Root-cause confidence:

- Aggregate trace confidence for the same root cause

Trace E, **Trace F** } Aggregate trace confidence → **Memory hardware fault: 0.92**

Rank	Root Cause	Aggregated score
1	Memory hardware fault	0.92
2	BootFlash Storage Failure	0.75
...	...	...

→

RCA report

Root cause: Memory hardware fault...

Detail analysis: ...

Action: ...

→ Operator

** Scoring and aggregation details are provided in our paper.*

Evaluation in production

> 1 Year

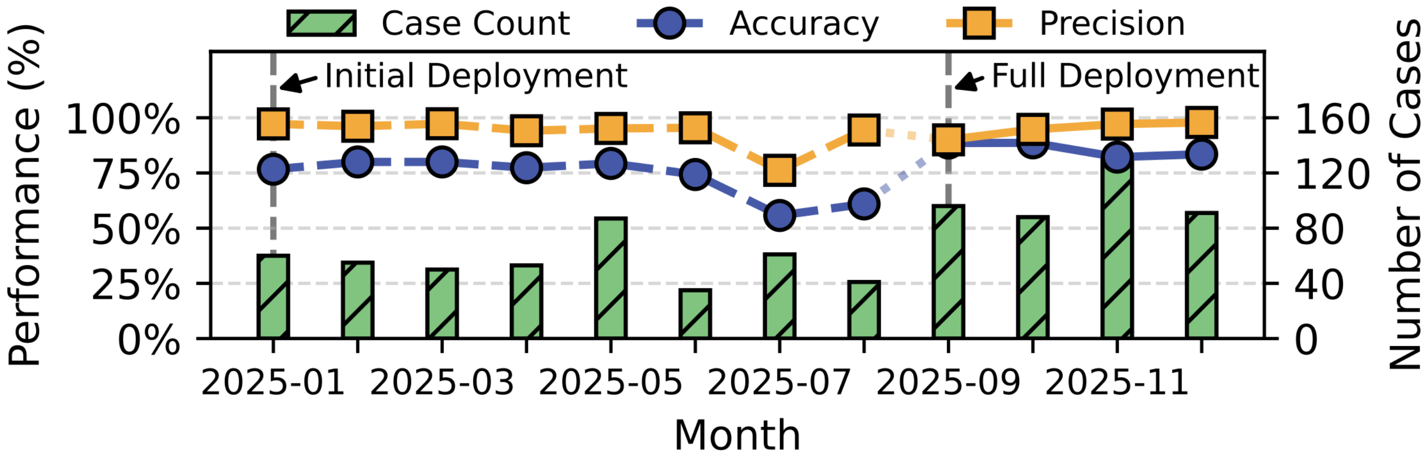
Deploy in production networks

846 Cases

Device-failure cases processed

101 cases/month

After full deployment



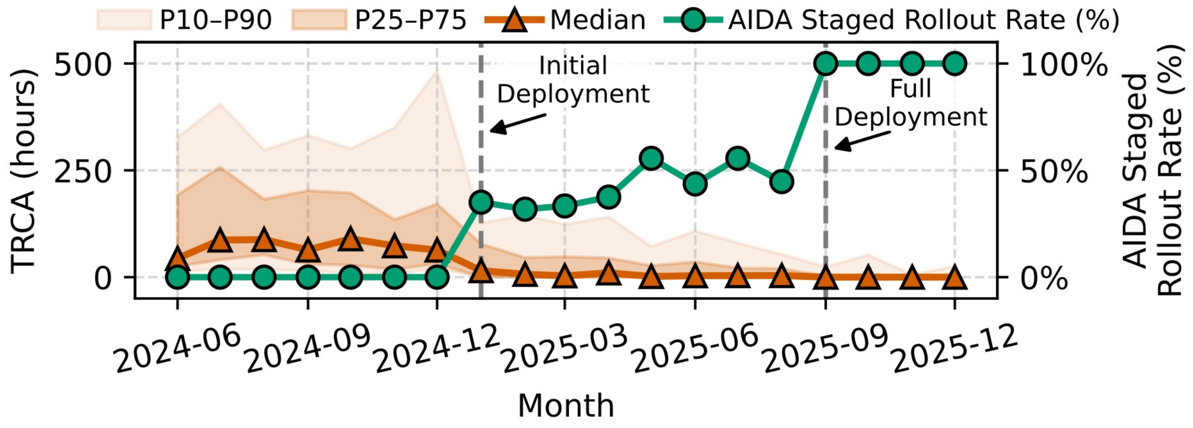
Full deployment

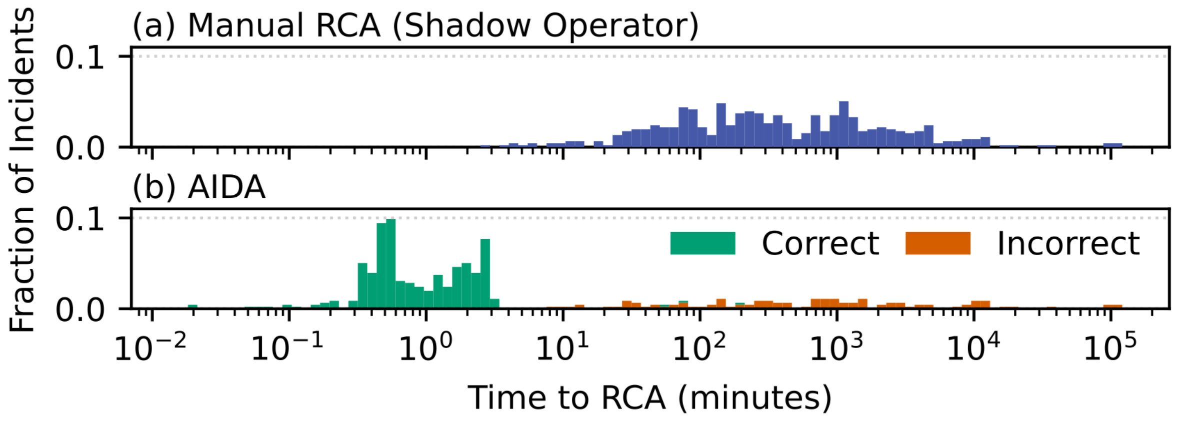
- **Average Accuracy: 85.4%**
  correct diagnoses / all cases
- **Average Precision: 95.4%**
  correct diagnoses / diagnosed cases

Precision-first diagnosis

- Uncertain cases → experts

RCA latency and automation scope





72.6h → 1.6min

Median time to RCA (TRCA)

329.9h → 19.4h

P90 long-tail diagnosis latency

Shadow evaluation: Parallel analysis by AIDA and experts

- Most cases within AIDA's coverage
- Complex long-tail cases remain with experts

Evaluation

Baselines

- **RCACopilot — Single-query email RAG**

  Combine **top-k emails from distinct error types** with the **log summary** → **one LLM diagnosis**

- **RCACopilot+ — Multi-query email RAG**

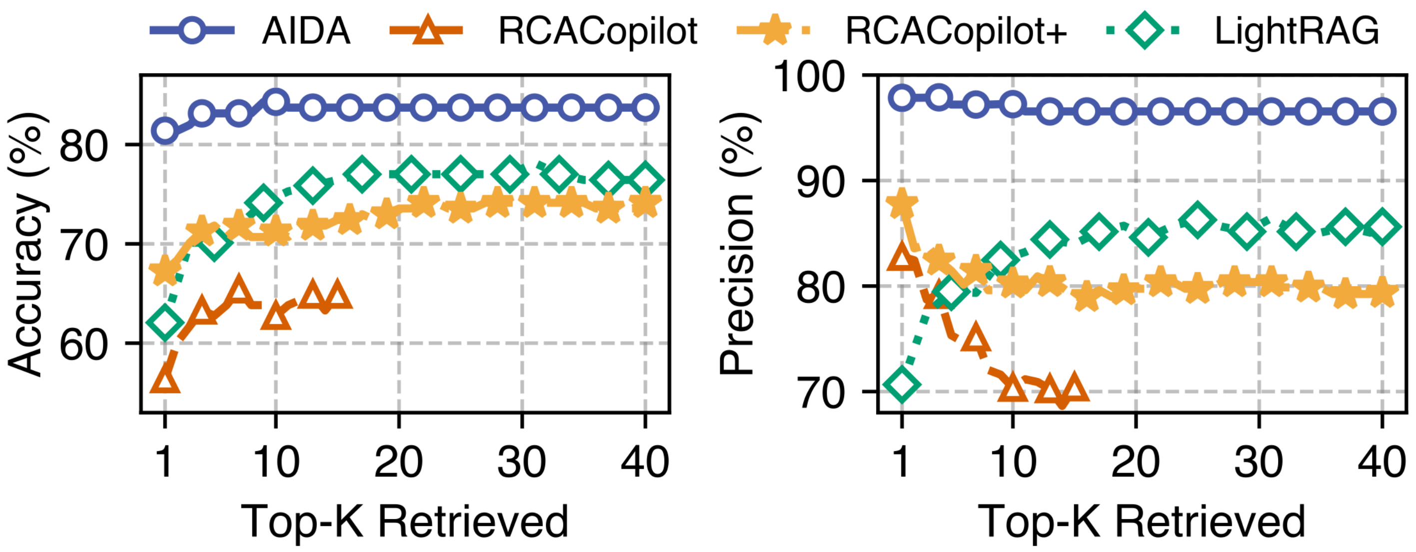
  Diagnose separately with each top-k email → **majority-voted diagnosis**

- **LightRAG — Generic graph RAG**

  Build a **generic KG** → perform **graph-based retrieval** → **one LLM diagnosis**

RCACopilot: Chen Y, et al., “Automatic root cause analysis via large language models for cloud incidents”, EuroSys ’24
LightRAG: Guo Z, et al., “LightRAG: Simple and Fast Retrieval-Augmented Generation”, EMNLP ’25

Evaluation



Why AIDA outperforms the baselines?

- **vs. RCACopilot:** **Reduces long-context distraction** and **covers diverse diagnostic paths** with structured reasoning chains
- **vs. RCACopilot+:** **Avoids error amplification** from majority voting through step-by-step evidence validation
- **vs. LightRAG:** **Reduces irrelevant retrieval** with an RCA-specific causal KG

Lessons learned

Evaluate alternatives systematically:

- Mitigate human bias with evidence-based comparison

Preserve diagnostic context:

- Differentiate similar log patterns by severity, frequency, magnitude, and timing

Monitor continuously:

- Distinguish transient software issues from persistent hardware failures through device-state tracking

Thank you!

Contact: Haoran Xu · xuhr6@mail2.sysu.edu.cn

Paper